CONTINUATION

CONTINUATION — HelixCode web-IDE platform

Revision: 4 · **Updated:** 2026-07-01 · **Status:** RELEASED **`codeserver-1.0.0-dev-0.0.3`** (real-account SSH-key auth) — live-validated (§11.4.169 matrix 23/23 PASS, 0 FAIL), tagged + pushed to all 4 mirrors, GitHub + GitLab releases published; stack LIVE at https://192.168.0.213:52443

Read FIRST on any fresh session: this file, then `git fetch --all`, then the auth-pivot spec + AUTH guide + feature ledger below. This is the §12.10 / §11.4.131 standing resumption anchor.

Current phase — `codeserver-1.0.0-dev-0.0.3` RELEASED (real-account SSH-key auth)

This round pivoted authentication to a **real-account, SSH-key challenge-response** model tying each HelixCode session to the real host user (`milosvasic`), host-native. Live-validated end-to-end on this host (full edge journey + §11.4.169 matrix **23/23 PASS, 0 FAIL** + Go gate **70 tests `-race`**), committed (`2746e0e`), tagged, pushed to all 4 mirrors (no force, §11.4.113), and released on GitHub + GitLab. The stack is LIVE + reachable for operator testing at **https://192.168.0.213:52443** — sign the `/login` challenge with an `~/.ssh/authorized_keys` key (no password). Non-blocking follow-ups: edge-container reboot-persistence; publicly-trusted Let's Encrypt (operator-gated); the loopback:8080 host-firewall recommendation.

- Changelog (this release): `docs/changelogs/codeserver-1.0.0-dev-0.0.3.md`
- Auth pivot spec (authoritative): `docs/superpowers/specs/2026-07-01-auth-pivot-ssh-key.md` (supersedes the live-PAM-login part of `docs/superpowers/specs/2026-07-01-real-account-code-server-design.md`)
- User guide: `docs/guides/AUTH.md`

- Feature ledger + validation verdicts: `docs/features/Status.md`
- Prior release: `docs/changelogs/codeserver-1.0.0-dev-0.0.2.md` (TLS / Let's Encrypt + §11.4.169 matrix)

Why the pivot (captured facts, §11.4.6)

This host is ALT Linux with the **tcb** shadow scheme, so a **non-root** service cannot verify passwords via PAM (`/etc/tcb/<user>/shadow` unreadable by the user; `tcb_chkpwd` setuid helper permission-denied to non-root ⇒ `pam_authenticate` returns `PAM_AUTH_ERR` for every password). A password gate would need root; the operator directive is **no sudo/root** and **all access via SSH keys**. `~/.ssh/authorized_keys` is readable and `ssh-keygen -Y sign`/`-Y verify` are supported, so SSH-key challenge-response runs fully **as non-root milosvasic**, with **no password and nothing stored**.

Architecture (this release)

```
Browser ──HTTPS──▶ Caddy (TLS edge, CONTAINERIZED rootless Podman;
HTTP/3 + brotli via custom xcaddy)
                    ├─ forward_auth ─▶ helix-auth (host-native
Gin, systemd --user, loopback:8081, NON-root)
                    └─ reverse_proxy ─▶ code-server (host-native
AS milosvasic, systemd --user,
                                        --auth none,
loopback:8080; inherits ~/.ssh, ~/.bashrc, host binaries)
```

- **code-server** — host-native `systemd --user`, `--auth none`, loopback:8080.
- **helix-auth** — host-native **Gin** gate, `systemd --user`, loopback:8081, non-root: SSH-key challenge (`GET /login`) → verify pasted signature vs `authorized_keys` (`ssh-keygen -Y verify`) → `__Host-` session cookie; `GET /auth` for forward_auth; **fails closed**.
- **Caddy** — sole containerized component: TLS edge, `forward_auth` (fail-closed), HTTP/3, `encode zstd br gzip`, `/proxy` 403, `X-Helix-User` strip.
- **Stack directive adopted:** Go + Gin + HTTP/3 (QUIC) + Brotli.

Feature status (§11.4.6 — honest)

The real-account + SSH-key rows in `docs/features/Status.md` are **In progress / PENDING validation** — no PASS is claimed until captured evidence lands. Testing: Go unit/integration/race (**70 tests, `-race`, 81.8% cover**) + shell suites `tests/types/{e2e,security,stress_chaos,concurrency,load,memory,benchmark,chall` `enges,helixqa}_auth.sh` + banks `tests/banks/helixcode-auth-{challenges,helixqa}.yaml`. The live aggregate is the conductor-filled placeholder in the changelog.

Honest boundaries carried into the release

- Editor "jail" is **cosmetic** (Explorer default only; terminal/Open-Folder/ extensions keep full host access by design).
- `--auth none` on loopback:8080 is a **residual** — host firewall rule for loopback:8080 recommended (all real auth is at the gate/edge).
- code-server pinned **4.117.0** (newest on npm at release time).
- Real public Let's Encrypt remains **operator-gated** (see `docs/guides/TLS.md`).

Migration

`scripts/install-auth.sh` (host-native, **no sudo**) provisions the `systemd --user` code-server + helix-auth units; `deploy/up.sh` brings up the Caddy edge. **`CODE_SERVER_PASSWORD` is retired** → `HELIX_AUTH_MODE=sshkey`, `HELIX_AUTH_ACCOUNT`, `HELIX_AUTH_AUTHORIZED_KEYS`, `HELIX_AUTH_PRINCIPAL`, `PROJECTS_ROOT` (no password parameter).

Immediate NEXT (to close the release)

1. Run the live §11.4.169 auth matrix against the freshly-installed host stack; capture evidence under `docs/qa/codeserver-1.0.0-dev-0.0.3/`.
2. Fill `LIVE VALIDATION AGGREGATE` in `docs/changelogs/codeserver-1.0.0-dev-0.0.3.md` with the real PASS/FAIL/SKIP counts (§11.4.6 — never invented).
3. Flip the `docs/features/Status.md` real-account rows to their verdicts once evidence lands.
4. Full-suite retest (§11.4.40) → tag `codeserver-1.0.0-dev-0.0.3` → publish to all upstreams via merge-onto-latest-main (no force-push, §11.4.113).

Binding constraints (every phase)

Port band **52000–52999**; publish only edge ports on `0.0.0.0`; **no secrets** (only git-ignored `.env`); **every gate paired with a §1.1 mutation**; **no `--force`/`--no-verify`/bypass**; hardlinked `.git` backup before destructive ops; **anti-bluff** (PASS needs captured runtime evidence, no invented numbers); stop + root-cause + full retest on any defect; release prefix `codeserver`; submodule commits propagate first; docs kept in sync.

Prior state (Phases 1–2, retained)

Phase 1 COMPLETE; Phase 2 core working+verified (Caddy edge + code-server stack on Podman; TLS1.3 edge on 52443, →301 on 52080; `$PROJECTS` bind-mounted). Owned submodules (8) at root carry the Constitution inheritance pointer; constitution pinned (see history). Prior release `codeserver-1.0.0-dev-0.0.2` shipped Let's Encrypt HTTPS (auto-renew + rotation) + the 14-suite §11.4.169 matrix.